

Atomic structure

Learning outcomes

By the end of this section you should be able to use the nuclear symbol ${}^A_Z\text{X}$ to deduce the number of protons, neutrons and electrons in atoms and ions.

The history of atomic theory has seen many developments over the past 100 years or so. However, none was more groundbreaking than the gold foil experiment carried out by Ernest Rutherford.

Experiments carried out in 1909 by Rutherford demonstrated that the atom is, in fact, largely empty space. In his experiment, helium nuclei (known as alpha particles) from a radioactive source were fired at thin sheets of gold foil. Their trajectories were detected to examine the structure of the atoms in the foil (**Figure 1**). Over 99% of the alpha particles passed straight through the metal foil, with a small percentage being deflected. The few large-angle deflections could be explained by the repulsion between the positively charged alpha particles and a strong positive charge localised in a small volume – in other words, the atomic nucleus. Rutherford determined that an atom is mostly empty space occupied by the negatively charged electrons, surrounding a very dense positively charged nucleus.

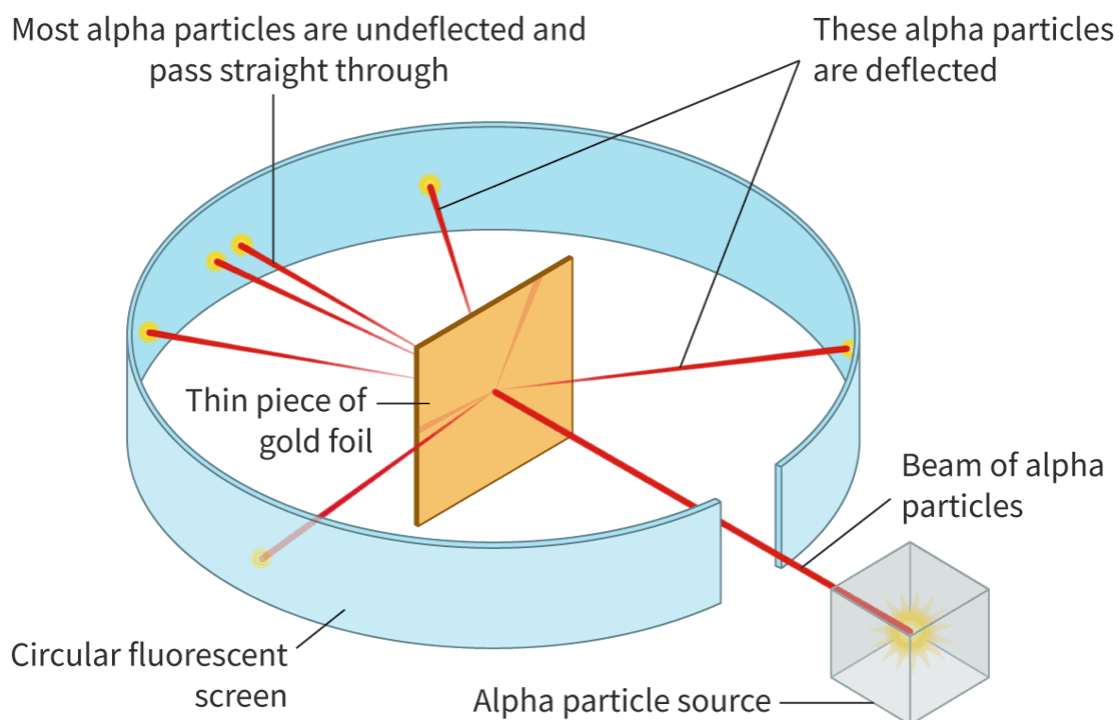


Figure 1. Thin piece of gold foil bombarded by alpha particles; note the deflection as some alpha particles collide with the positively charged nucleus.

 More information for figure 1

The image is a diagram illustrating an experiment where alpha particles are being fired at a thin piece of gold foil. It shows a circular fluorescent screen with a small gap where a cube labeled as an "alpha particle source" is located. A beam of alpha particles is emitted from the source and directed towards the center of the circle, where it hits a square labeled "thin piece of gold foil." Some of the alpha particles pass straight through the foil and reach the circular screen, as indicated by lines labeled "most alpha particles are undeflected and pass straight through." Meanwhile, other particles are deflected, shown by diverging lines labeled "these alpha particles are deflected." This diagram represents the Rutherford experiment that helped demonstrate the structure of the atom.

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In section S1.1.1.a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>), you learned that elements are pure substances and are made up of only one type of atom. Atoms are the smallest particles that retain the properties of the element. A copper atom, for example, will have the same properties as a larger piece of copper, consisting of millions of copper atoms. You can think of an atom as having two main regions: the nucleus and the space around the nucleus where the electrons are located. The arrangement of the electrons is covered in more detail in subtopic S1.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/>).

The nucleus, which is located at the centre of the atom, contains the protons and the neutrons (collectively known as nucleons). The nucleus of an atom is extremely dense, somewhere in the region of $2.3 \times 10^{17} \text{ kg m}^{-3}$, as it contains almost all of the mass of the atom. The nucleus is positively charged because it contains protons, which have a positive charge and neutrons, which are neutral. An atom of carbon-12 is shown in **Figure 2**.

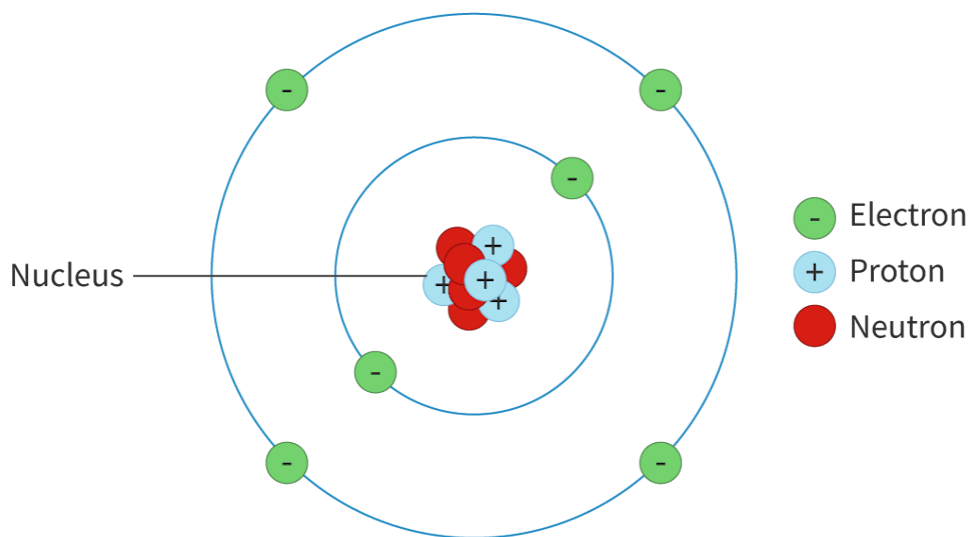


Figure 2. An atom of carbon-12 showing the location of the subatomic particles.

 More information for figure 2

The diagram illustrates the structure of a carbon-12 atom. In the center, there is an inner circle representing the nucleus, which contains eight overlapping circles. Four of these circles have plus signs, labeled as protons, and the other four are blank and colored red, labeled as neutrons. Surrounding the nucleus are two orbitals. The first inner orbital contains two small circles with minus signs, representing electrons. The outer orbital has four small circles, also with minus signs, indicating more electrons. The diagram is labeled to specify the electrons, protons, and neutrons, and a line points to the nucleus highlighting its location in the atom.

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Theory of Knowledge

Subatomic particles cannot be viewed with the limitations of current technology. What role does faith play in your understanding of their existence?

The subatomic particles have specific masses and charges. Protons are positively charged and have a relative mass of 1. Neutrons have no charge (they are neutral) and also have a relative mass of one. Electrons are negatively charged and have a relative mass of approximately $\frac{1}{2000}$ th the mass of a proton, which is considered to be negligible and is usually not included when calculating the mass of an atom. The location, mass and charges of the subatomic particles are summarised in **Table 1**.

Table 1. Relative mass and charge of the subatomic particles

Subatomic particle	Symbol	Relative charge	Relative mass	
Proton	p^+ or ${}_1^1p$	+1	1	
Neutron	n^0 or ${}_0^1n$	0	1	
Electron	e^- or ${}_1^0e$	-1	$\frac{1}{2000}$	In energy

Relative masses and charges are used for the subatomic particles because the actual values are so small. For example, a proton has a mass of 1.67×10^{-27} kg which is a very small mass indeed. Now you can see that it is much easier to use the relative mass of 1 for both the proton and the neutron. Atoms are also extremely small. An aluminium atom, for example, has a radius of 1.24×10^{-11} metres. Based on this size, a thin sheet of aluminium foil is approximately 100 000 atoms thick.

Nature of Science

Aspect: Falsification

Scientists can use evidence to falsify a theory, but they cannot prove with certainty that a theory is true. In fact, all scientific claims must be expressed so that they can be proved false with new evidence.

Atomic theory in the 19th century proposed a solid 'plum pudding' model of an atom. In the 20th century, this theory was falsified by new evidence that showed that an atom was not solid at all, but mostly empty space!

Atoms are electrically neutral because they contain the same number of protons and electrons. An atom of carbon, for example, contains six protons and six electrons. Because there are equal numbers of each particle, the opposite charges cancel out. Atoms of the same element can have different numbers of neutrons and are known as isotopes.

Making connections

Subtopic S1.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43119/>), covers the factors that affect the chemical properties of an atom.

Nuclear symbol notation

The identity of an atom is determined by the number of protons in its nucleus which is represented by the atomic number. The atomic number (symbol Z) is the number of protons in the nucleus of an atom. The mass number (symbol A), also called the nucleon number, is the number of protons and the number of neutrons in the nucleus of an atom. The nuclear symbol notation, or the nuclear symbol, is used to represent the symbol of the element (X), together with its atomic number (Z) and its mass number (A).

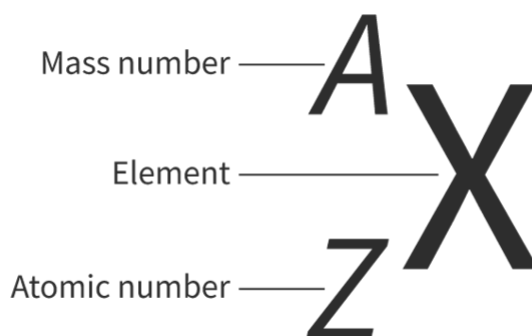


Figure 3. The nuclear symbol of an element, X showing the mass number (A) and the atomic number (Z).

 More information for figure 3

The diagram illustrates the nuclear symbol of an element represented by a large letter 'X.' At the top right of the 'X' is an italicized letter 'A,' which is labeled as the mass number. Below 'X,' there is an italicized letter 'Z,' labeled as the atomic number. Arrows point from the labels "Mass number," "Element," and "Atomic number" to the letters 'A,' 'X,' and 'Z,' respectively, indicating their corresponding positions and meanings within the nuclear symbol structure. This explanation aids in understanding how the mass number and atomic number are represented in the nuclear notation of an element.

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Note that the atomic number is often omitted and can be deduced by the symbol of the element. So the nuclear symbol for the isotope iron-54 can be shown as ${}^{54}_{26}\text{Fe}$ or ${}^{54}\text{Fe}$.

Study skills

A copy of the periodic table can be found in [section 7](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) of the DP chemistry data booklet. The atomic number, element symbol and relative atomic mass can be found in this section. You will notice that your periodic table contains non-whole number values for relative atomic mass. You will learn more about this in [section S1.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/>). The names of the elements, together with their symbols, can be found in [section 6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/names-of-the-elements-id-45108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/names-of-the-elements-id-45108/>) of the DP chemistry data booklet.

Using the nuclear symbol notation, the number of protons, neutrons and electrons in an atom (or ion) can be determined. For example, in ${}^{54}_{26}\text{Fe}$ we can determine the atomic number of iron is 26 and its mass number is 54. The atomic number tells us the number of protons in the nucleus, so ${}^{54}_{26}\text{Fe}$ has 26 protons in its nucleus. The mass number is 54 which is the number of protons plus the number of neutrons in the nucleus. To find the number of neutrons, we subtract the atomic number from the mass number. So ${}^{54}_{26}\text{Fe}$ has 28 neutrons in its nucleus ($54 - 26 = 28$).

Worked example 1

Determine the number of protons, neutrons and electrons in the isotope ${}^{16}_8\text{O}$.

The atomic number is 8 and the mass number is 16, so there are 8 protons, 8 neutrons and 8 electrons in this atom.

Because this is an atom, the number of protons is equal to the number of electrons.



Exercise 1

Click a question to answer



1. Determine the number of protons, neutrons and electrons in the isotope ${}_{19}^{39}\text{K}$.



Check answers

Worked example 2

Write the nuclear notation for the isotope lithium-8.

The mass number of the isotope is 8 and the atomic number is 3. The symbol for lithium is Li. The nuclear notation is therefore ${}^8_3\text{Li}$.



Exercise 2

Click a question to answer



1. True or false?



The nuclear notation for the isotope sulfur-34 is ${}^{34}_{16}\text{S}$.

Check answers

Nuclear notation of ions

The nuclear symbol notation can also be used to represent ions. Ions are formed when atoms either lose or gain electrons. Recall that atoms are neutral because they contain equal numbers of protons and electrons. Because the electrons are located outside of the nucleus they can be lost or gained relatively easily. The

formation of ions will be covered in more detail in [section S2.1.1](#)

([https://app.kognity.com/study/app/preview-p/sid-426-cid-](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/)

[753301/book/formation-of-ions-id-43120/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/)).

- Positive ions are formed when atoms lose electrons, Sodium, for example, loses one electron to form a 1+ ion. The nuclear symbol for the sodium ion is represented as ${}_{11}^{23}\text{Na}^{+}$. Note the positive sign which shows the ion has a 1+ charge. This sodium ion has 11 protons, 12 neutrons and 10 electrons. To determine the number of electrons, we subtract one from the number of protons because it is a 1+ ion.
- Negative ions are formed when atoms gain electrons. Fluorine can gain one electron to form a 1- ion with the nuclear symbol ${}_{9}^{19}\text{F}^{-}$. This ion has 9 protons, 10 neutrons and 10 electrons. To determine the number of electrons, we add one to the number of protons because it is a 1- ion.

Worked example 3

Determine the number of subatomic particles in the ${}_{19}^{39}\text{K}^{+}$ ion.

The atomic number is 19 and the mass number is 39, so there are 19 protons and 20 neutrons. A potassium atom has 19 electrons but because it is a 1+ ion, subtract one from the number of electrons, which gives 18 electrons.



Exercise 3

Click a question to answer



1. Determine the number of subatomic particles in the ${}_{16}^{34}\text{S}^{2-}$ ion.



Check answers

Worked example 4

Deduce the nuclear notation for the ion with 7 protons, 7 neutrons and 10 electrons.

The ion has 7 protons so the atomic number is 7. It also has 7 neutrons so the mass number is 14. It has 3 more electrons than protons so the charge on the ion is 3⁻. Therefore, the nuclear notation is ${}^{14}_{7}\text{N}^{3-}$.



Exercise 4

Click a question to answer



1. The ion ${}^{25}_{12}\text{Mg}^{2+}$ contains _____ protons,
_____ neutrons and _____ electrons.

Check answers

Use the activity in the next section (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/activity-build-an-atom-id-43167/>) to build an atom.